

Area Models for Two-Digit Multiplication

Answer Key

Grades 4–5 Mathematics

Quick Answers

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|---|---|
| 1. $10 \times 20 = 200$, $10 \times 3 = 30$, $4 \times 20 = 80$, $4 \times 3 = 12$, the total $14 \times 23 = 322$ | 6. the estimate = 800, the exact product = 798 |
| 2. $20 \times 30 = 600$, $20 \times 1 = 20$, $6 \times 30 = 180$, $6 \times 1 = 6$, the total $26 \times 31 = 806$ | 7. the unknown side = 24, check with the bottom cell = 24 |
| 3. $47 = 40 + ? = 7$, $35 = 30 + ? = 5$ | 8. 884 |
| 4. the product = 360, — | 9. the correct product = 990, — |
| 5. $30 \times 20 = 600$, $30 \times 8 = 240$, $2 \times 20 = 40$, $2 \times 8 = 16$, the total $32 \times 28 = 896$ | 10. $60 \times 40 = 2400$, $60 \times 7 = 420$, $3 \times 40 = 120$, $3 \times 7 = 21$, the total $63 \times 47 = 2961$ |
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What is verified

8 of 10 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 4, 9. The worked solution states what a correct response must contain.

Curriculum

Level: Grades 4–5 Mathematics

4.NBT.B.5 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–4 of 5 across 31 tagged checks.

Common wrong answers

9. *If they got 900: multiplied 45 by 20 only and dropped the 45 x 2 cell of the model*

1. 14×23 with the model split $10 + 4$ and $20 + 3$.

Solution: Each cell multiplies one part of each factor: $10 \times 20 = \boxed{200}$, $10 \times 3 = \boxed{30}$, $4 \times 20 = \boxed{80}$, $4 \times 3 = \boxed{12}$.

Add the cells: $200 + 30 + 80 + 12 = 322$, so $14 \times 23 = \boxed{322}$

2. 26×31 with the model split $20 + 6$ and $30 + 1$.

Solution: $20 \times 30 = \boxed{600}$, $20 \times 1 = \boxed{20}$, $6 \times 30 = \boxed{180}$, $6 \times 1 = \boxed{6}$.

Add: $600 + 20 + 180 + 6 = 806$, so $26 \times 31 = \boxed{806}$

3. Fill in the splits for 47×35 .

Solution: A two-digit number splits into its tens and its ones: $47 = 40 + 7$, so the first blank is

7

. Likewise $35 = 30 + 5$:

5

4. Label the model for 24×15 and find the product.

Solution: Split $24 = 20 + 4$ and $15 = 10 + 5$. The four cells are $20 \times 10 = 200$, $20 \times 5 = 100$, $4 \times 10 = 40$, $4 \times 5 = 20$. Sum: $200 + 100 + 40 + 20 = 360$, so $24 \times 15 =$

360

A correctly filled model shows exactly those four cells.

5. 32×28 : the four cells and their total.

Solution: Split $32 = 30 + 2$ and $28 = 20 + 8$: $30 \times 20 =$

600

, $30 \times 8 =$

240

, $2 \times 20 =$

40

, $2 \times 8 =$

16

.

Total: $600 + 240 + 40 + 16 = 896$, so $32 \times 28 =$

896

6. Estimate 38×21 by rounding, then compute exactly.

Solution: Round each factor to the nearest ten: 38 rounds to 40 and 21 rounds to 20, so the estimate is $40 \times 20 =$

800

Exact, by the model ($30 + 8$ and $20 + 1$): $600 + 30 + 160 + 8 = 798$: $38 \times 21 =$

798

 — close to the estimate, so it is reasonable.

7. Rows 10 and 3; cells 240 and 72; unknown shared side.

Solution: The top cell says $10 \times \text{side} = 240$, so the side is $240 \div 10 = 24$.

24

Check with the bottom cell: $72 \div 3 =$

24

 — both cells agree, so the side is 24.

8. A patio is 34 tiles long and 26 tiles wide. How many tiles?

Solution: Split $34 = 30 + 4$ and $26 = 20 + 6$. Cells: $30 \times 20 = 600$, $30 \times 6 = 180$, $4 \times 20 = 80$, $4 \times 6 = 24$. Sum: $600 + 180 + 80 + 24 = 884$ tiles. $34 \times 26 =$

884

9. Elena computes 45×22 and writes 900.

Solution: Split $22 = 20 + 2$. The model has two cells: $45 \times 20 = 900$ and $45 \times 2 = 90$. Elena stopped after the first cell. The full product is $900 + 90 = 990$:

990

Full-credit explanation: the model for $\times 22$ has a tens cell *and* a ones cell; Elena dropped the 45×2 cell.

10. Challenge: 63×47 .

Solution: Split $63 = 60 + 3$ and $47 = 40 + 7$: $60 \times 40 =$

2400

, $60 \times 7 =$

420

, $3 \times 40 =$

120

, $3 \times 7 =$

21

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Total: $2400 + 420 + 120 + 21 = 2961$, so $63 \times 47 =$

2961